

Empirical Analysis of Network Structure Across Twitter Interaction Types

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Overview

Our project is an empirical analysis of the SNAP Higgs Twitter dataset. We compare four different Twitter networks: the follower (social) network, retweet network, reply network, and mention network. Each network is represented as a directed weightless graph where users are the nodes and the interactions are the edges. The goal of our project is to understand how different types of relationships create different network structures.

First, we use in-degree analysis to measure how attention is distributed across each network. We also do a BFS random pair analysis experiment to estimate how connected each graph is and how long shortest paths between reachable users are. We also compute clustering coefficients to compare how much local community structure exists in each graph.

Data Set

Link: <https://snap.stanford.edu/data/higgs-twitter.html>

We used the mention, reply, social, and retweet graph.

Hypotheses

In-Degree Concentration: We expect the retweet and mention graph to show stronger in-degree concentration than the reply graph. This hypothesis is due to the idea that retweets and mentions are often given to popular accounts, so we expect a small number of users to receive a large number of retweets and mentions. As for replies, they tend to be more conversational and could be more spread out and would have lower maximum in-degree and less extreme concentrations.

BFS Connectedness and Shortest Paths: We hypothesize that the follower graph will be the most connected network because following is a broader and lower-effort relationship unlike replies or mentions. A user can follow many accounts without directly interacting with them, so the follower graph should contain more paths between random users. Consequently, the average shortest paths between random users will be significantly less than the other networks.

Clustering Coefficient: We hypothesize that the follower graph would have the highest clustering coefficient because users who follow one another online are more likely to be friends in real life and exhibit triadic closure. We also expect the reply graph to have the lowest clustering coefficient, since replies are often one-to-one interactions rather than interactions occurring within larger groups.

Analysis of Results

1. In-Degree Analysis

Method

In-degree measures how many edges point into a user. In the followers graph, high in-degree means many users follow that account. In the retweet graph, high in-degree means many users retweeted that account. And a similar idea applies to the reply and mention graph. For each network, we computed the average in-degree and max in-degree and plotted the distribution for each network.

Results

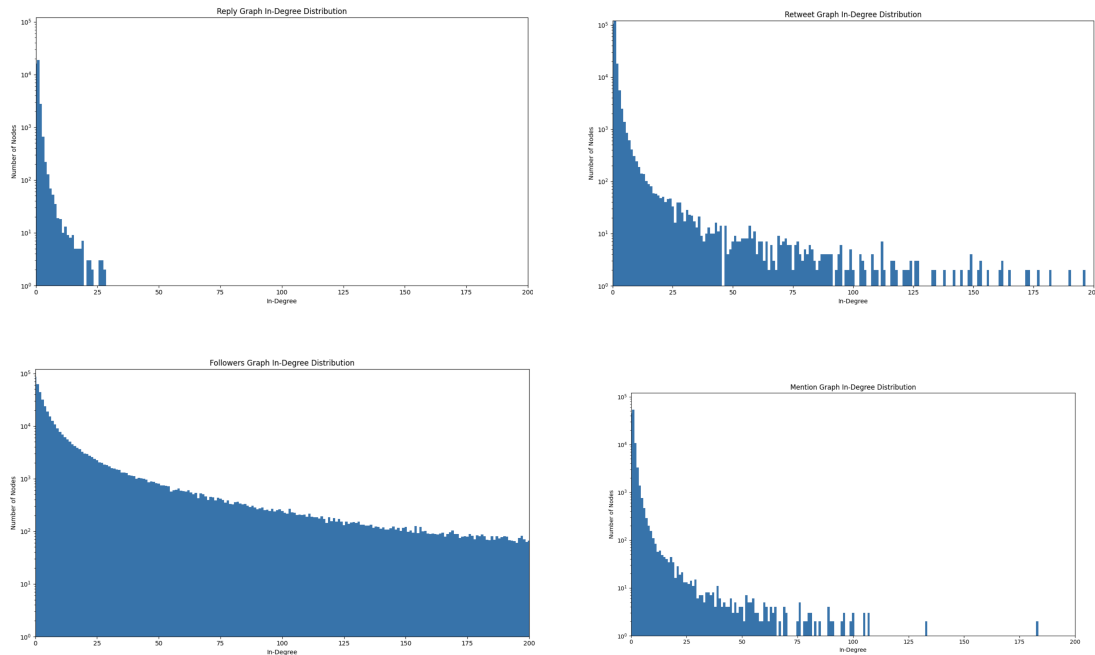
Graph	Nodes	Edges	Average In-Degree	Max In-Degree
Mention	116,408	150,818	1.2956	11,953
Social/Follower	456,626	14,855,842	32.5339	51,386
Retweet	256,491	328,132	1.2793	14,060
Reply	38,918	32,523	0.8357	1,206

The above table shows that the follower graph is much denser than the active interaction graphs, with an average in-degree of 32.53 while the mention, retweet, and reply graphs have averages close to or below 1.3. This makes sense because following is low-effort and long term, while retweeting, replying or mentioning need active engagement. The retweet and mentions graph also show strong concentration. Although their average in-degrees are low, their maximum in degrees are very high. The most retweeted user has an in-degree of 14060 and the most mentioned user has an in-degree of 11953. The reply graph is much smaller and has the lowest degree at 1206. This supports the idea that replies are less centralized than retweets and mentions, as they appear to represent more direct conversation, while retweets and mentions are more tied to public visibility.

Graphs

The shapes of the in-degree distribution graphs support the concentration ratios. The follower graph has a smoother downward curve, showing a more continuous spread of users across different in-degree values; although some users have many more followers than others, the distribution is less sharply broken up. The retweet graph drops off quickly at first, then has many scattered bars at the high end, meaning most users receive very few retweets while a small number receive unusually many. The reply graph has almost no scattered high-end bars,

matching its lower maximum in-degree and suggesting that replies are less dominated by a few central users. The mention graph is initially smooth but still has a few scattered high-end bars, showing that most users are rarely mentioned while a few highly visible users receive far more mentions than everyone else.



Conclusions

The in-degree analysis supports Hypothesis 1. Retweets and mentions are much more concentrated than replies, while the follower graph is much denser overall. This shows that different Twitter relationships capture different kinds of attention: followers measure broad popularity, retweets measure amplification, mentions measure visibility, and replies measure direct engagement.

2. Connectivity and Shortest Path Analysis (BFS)

Method

To test whether the follower network is more connected than the other networks, we ran a random-pair BFS experiment on each of the graphs. We did this by randomly sampling distinct ordered node pairs, then ran the BFS algorithm to obtain the shortest directed path from one node to the other.

Results

We collected the results in the following table:

Graph	Trials	Connected Pairs	Fraction Connected	Average Shortest Path Among Connected Pairs
Mentions Graph	10,000	109	0.0109	7.9266
Retweet Graph	10,000	25	0.0025	8.5600
Reply Graph	10,000	19	0.0019	10.9474
Social Graph	100	80	0.8000	4.6750

There are 2 things to note.

1. The follower (social) graph connects 80% of the random pairs, while the other graphs connect 0.2-1.1% of random pairs.
2. The follower graph has the smallest average shortest path among the graphs. The follower graph has an average shortest path of 4.68, while the other graphs have larger values ranging from 7.93 to 10.95.

Additionally, it's important to note that, for the follower graph, we only did 100 trials instead of 10,000 as done for the other graphs. Nevertheless, it still dominates by orders of magnitude.

Conclusions

Our hypothesis is supported by both evaluations. The follower graph is much more connected than the other networks and has the shortest average path length among connected pairs. This holds despite the reduced trial count, which can only depress the follower graph's measured performance. This makes sense because following is easy and low-effort, so users build up lots of connections, while mentions, retweets, and replies take more effort and tend to cluster around a few popular accounts. Overall, easy connections like follows create well-connected and denser graphs, while interaction-based edges create sparser ones.

3. Clustering Coefficient Analysis

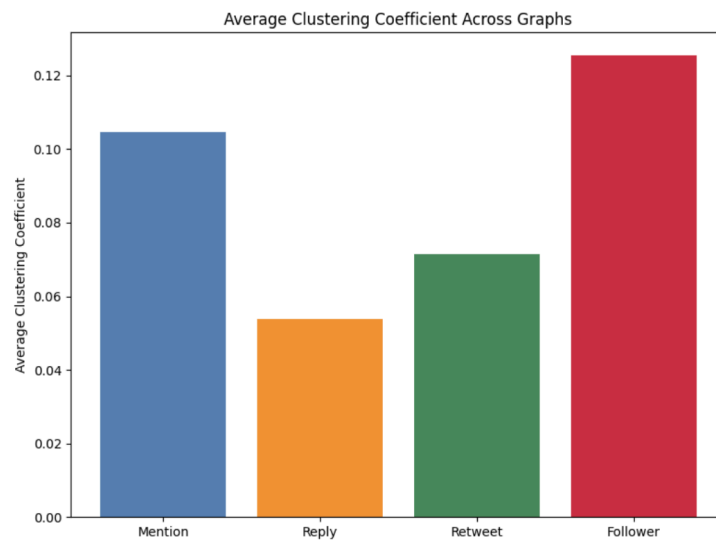
Methods

To estimate the overall clustering coefficient of each network efficiently, we randomly sampled 10,000 nodes from each graph, computed their clustering coefficients, and averaged the results. Because the Higgs Twitter networks are directed graphs, the cluster coefficient was computed as the proportion of directed links that exist between nodes in its neighborhood (including the in- and out-neighborhood) out of all the directed links that could possibly exist between them.

Results

The follower graph produced the highest average clustering coefficient among the four graphs (0.125). This is to be expected since the follower graph directly represents social relationships, making triadic closure much more likely to occur. Additionally, the clustering coefficient for mentions wasn't much lower (0.104), suggesting that mentions also tend to occur within connected communities. This is likely because users frequently mention people who are already part of the same social circle (like in the same photo) leading to many interconnected triangles within the graph. Finally, the retweets and replies had the lowest clustering coefficient (0.071 and 0.054), indicating that these interactions are less likely to occur within tightly connected groups. Retweets often spread information broadly across different parts of the network rather than within close communities, while replies are frequently direct one-to-one interactions that do not necessarily create additional connections between neighboring users.

Graph



Conclusions

The clustering coefficient analysis supports our hypothesis that the follower graph would exhibit the strongest local community structure. The results also show that mentions frequently occur within connected social groups, while retweets and replies are less community-oriented interactions. Overall, the analysis demonstrated that different types of interactions noticeably different network structure and levels of triadic closure.